



COURSE PACK

Course pack and syllabus

Introduction to Digital Forensics

A 14-week undergraduate course: Windows artifacts, memory, network and mobile evidence, ending in a graded capstone.

Length 14 weeks

Discipline Digital Forensics

Level Undergraduate, second or third year

Prerequisites Operating systems fundamentals and basic networking. No prior forensics.

Prepared 2026-10-07

Every lab in this course generates a different variant for each student from one case, with an answer key verified against that student's own evidence. Grading and the student portal run on the instructor's machine.

Learning outcomes

1. Acquire and verify evidence, and keep a chain of custody.
2. Recover program execution, file, USB and persistence artifacts from a Windows system.
3. Build and defend a timeline from several artifacts.
4. Triage memory, network and mobile evidence with free tools.
5. Write an investigation report that states findings, cites artifacts and explains method.

Assessment

Weekly labs 50% (best 10 of 12), midterm practical 20%, capstone practical 20%, investigation report 10%.

Tools

Autopsy, Registry Explorer, EvtxECmd, PECmd, LECmd, JLECmd, Volatility 3, Wireshark, DB Browser for SQLite, Timesketch. All free; one-run install scripts are on the Tools page.

Week by week

Wk	Topic	Objectives	Case and mode	Assessment
1	Evidence handling and the registry	Explain acquisition, hashing and chain of custody. Read USBSTOR and device properties in a SYSTEM hive.	First Look: USB Device History (practice)	Lab 1
2	Program execution	Interpret Prefetch, UserAssist and Amcache. Tell a decoy execution from the real one.	First Look: What Ran on This Machine (practice)	Lab 2
3	Files the user touched	Parse LNK files and jump lists. Recover a target path and its timestamps.	First Look: Recent Files and Jump Lists (practice)	Lab 3
4	Persistence	Enumerate Run keys, services and scheduled tasks. Rank persistence mechanisms by stealth.	First Look: What Runs at Startup (practice)	Lab 4
5	Windows event logs	Read 4720, 4732, 7045 and 1102 events. Distinguish the acting account from the target account.	First Look: Accounts and Services in the Logs (practice)	Lab 5
6	Timelines	Normalise timestamps across artifacts. Build and present a super-timeline.	First Look: Building a Timeline (practice)	Lab 6

Wk	Topic	Objectives	Case and mode	Assessment
7	Midterm practical	Apply weeks 1 to 6 to a full insider-theft case under time.	Insider Data Theft via USB (timed exam, 75 min)	Midterm 20%
8	Browser and download forensics	Trace a download from history to disk. Read browser SQLite stores.	First Look: Suspicious Download (practice)	Lab 7
9	Cloud exfiltration	Correlate browser, LNK and email evidence. Prove which file left and when.	Exfiltration to a Personal Cloud Drive (practice)	Lab 8
10	Memory forensics	Run Volatility 3 process, network and DLL plugins. Identify the suspicious process.	First Look: What's in Memory (practice)	Lab 9
11	Code injection in memory	Find injected and unlinked modules. Tie a memory finding to disk and network artifacts.	Injected Code in Memory (practice)	Lab 10
12	Network forensics	Read a PCAP in Wireshark. Extract hosts, sessions and a transferred file.	First Look: Reading a Packet Capture (practice)	Lab 11
13	Mobile forensics	Query sms.db and recover deleted rows. Convert Cocoa time.	The Texts They Deleted (practice)	Lab 12
14	Capstone practical and report	Work a multi-host intrusion end to end. Deliver an investigation report with chain of custody.	Credential Theft and Lateral Movement (timed exam, 120 min, report)	Capstone 20% + report 10%

Running the course

In Red Herrings, open Courses, pick this course, enter the class size or roster and press Generate the whole course. One job per week is created with the mode and time limit above; hand out each week's zips or portal links when that week arrives. Scores export to your LMS from each job.